



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2011 - 127



RJR-AGRO Mini Rice Thresher
Model: 101311

Test Report No.	:	2011 - 127
Date of Test	:	October 13, 2011
Test Valid Until	:	October 13, 2016
Brand	:	RJR-AGRO
Model	:	MT101311
Type	:	Open, single cylinder, with cleaning component (blower only)
Manufacturer	:	RJR AGRO-INDUSTRIAL ENTERPRISES Bangcud, Malaybalay City Bukidnon
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The RJR-AGRO MT101311 Mini Rice Thresher powered by a 5.97 kW KUBOTA RK80-1T diesel engine was tested for performance using an unknown rice variety at 18.2% MC_{wb}, Grain-Straw Ratio of 0.62, and Straw Length of 539 mm, the thresher had a Threshing Capacity of 1024 kg/h with threshed grain Purity of 94.52%. At a Threshing Cylinder Speed of 600 rpm and Peripheral Speed of 954 m/min, Threshing Recovery was 99.71% and Threshing Efficiency was 100%. The machine had a Total Threshing Loss of 0.29% consisting of 0.12% Blower Loss, 0.17% Separation Loss, no Unthreshed Loss and no Scattering Loss.

For purposes of standardization, the average Corrected Threshing Capacity was 877.4 kg/h based on 20% MC_{wb}, 0.55 Grain-Straw Ratio and 100% Purity.

Grain quality analysis of threshed grains showed a net percent Cracked Grain of 4.34 % and no Broken Grains.

The Fuel Consumption Rate of the 5.97 kW KUBOTA RK80-1T diesel engine was 1.57 L/h.

During the test, six persons were needed to operate the thresher: two for hauling of test materials to the thresher; one for feeding of test materials; one for collecting of clean threshed grains; and, two for bagging.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance using KUBOTA RK80-1T diesel engine as its primemover. The test was conducted in Brgy. Colonia, Valencia City, Bukidnon on October 13, 2011 and was evaluated on the following items:

1. Threshing Capacity
2. Quality of Work
 - a. Purity of Threshed Grain
 - b. Percent Cracked Grain
 - c. Percent Broken Grain
3. Threshing Recovery
4. Threshing Efficiency
5. Threshing Losses
 - a. Blower Loss
 - b. Separation Loss
 - c. Unthreshed Loss
 - d. Scattering Loss

METHOD OF TEST

The thresher was tested based on the Philippine Agricultural Engineering Standard “PAES 205:2000 Agricultural Machinery – Mechanical Rice Thresher - Methods of Test”.

OPERATOR'S MANUAL

No operator's manual nor brochure of the thresher was submitted.

DESCRIPTION OF THE MACHINE

The RJR-AGRO MT101311 Mini Rice Thresher as shown in **Figure 1**, was basically an axial flow type mechanical paddy thresher with cleaning component (blower only). It was powered by a 5.97 kW KUBOTA RK80-1T diesel engine which drove directly the threshing cylinder through a double groove B-section pulley and V-belt. The threshing cylinder in turn drove the fan shaft through a single groove B-section pulley and V-belt. Belt power transmission system of the rice thresher is shown in **Figure 2**.

The machine consisted of the following main components: feeding table, threshing assembly and fan. The folding type feeding table (**Fig. 3**) was attached to the main frame through bolts and nuts. Feeding was controlled manually.

The threshing assembly (**Fig. 4**) consisted of a single, open type cylinder and a lower concave grill. The threshing cylinder had a length of 710 mm and a diameter of 506 mm and had eight slanted anchor bars. Each anchor bar was bolted with eight pegteeth which were arranged offset with respect to the pegteeth in the next anchor bar. The lower concave grills were made of 12 mm flat bar with 6 mm thickness. The threshing section was provided with a top cover hinged to the main frame. The top cover was welded with six spiral louvers.

The cleaning assembly consisted of a straight radial type centrifugal fan (**Fig. 5**) which separated the straw chaffs and other light impurities from the threshed grain which fall below the threshing cylinder. Clean threshed grains were collected manually for bagging.

The main frame of the machine was made of hollow square bar.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. RJR-AGRO Mini Rice Thresher
Model: MT101311



Figure 2. Belt-Pulley Power Transmission System



Figure 3. Feeding Table

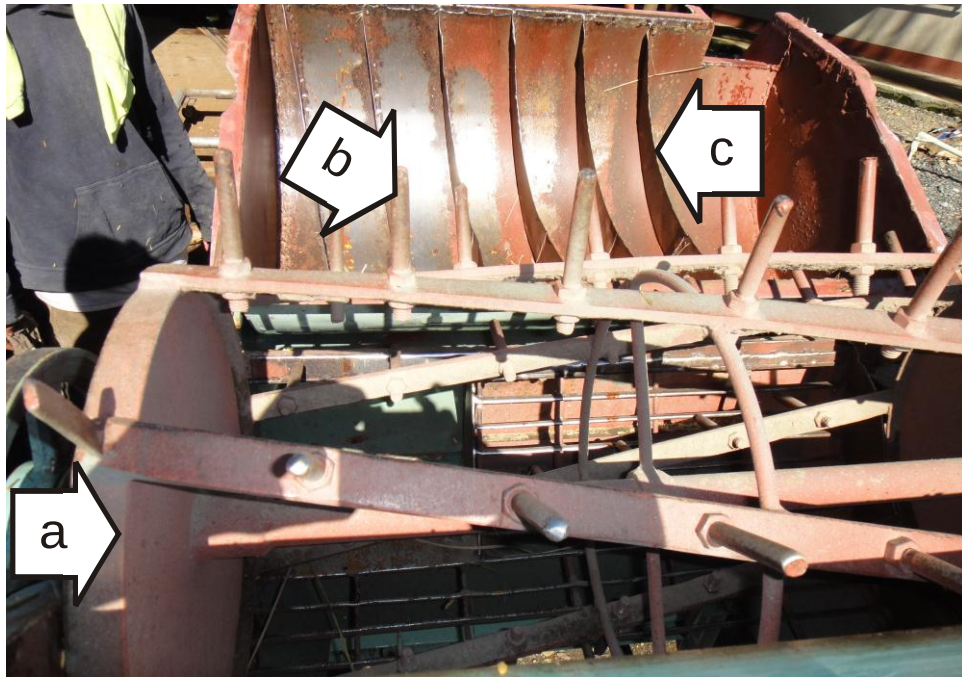


Figure 4. Threshing Assembly
a) Threshing cylinder
b) Pegteeth
c) Louver

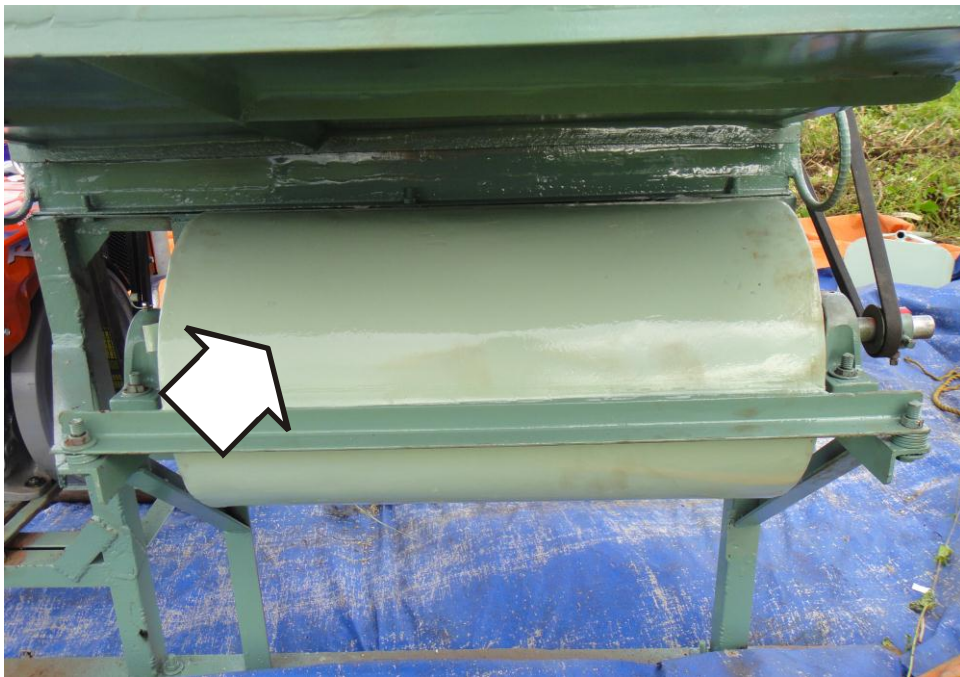


Figure 5. Straight Radial Centrifugal Fan

SPECIFICATIONS

Table 1. Detailed specifications of the RJR-AGRO MT101311 Rice Thresher

A.	Brand	:	RJR-AGRO
B.	Model	:	MT101311
C.	Serial No.	:	-
D.	Manufacturer	:	RJR AGRO-INDUSTRIAL ENT. Malaybalay City, Bukidnon
E.	Overall Dimensions (Transport mode) and Weight		
1.	Length (mm)	:	1460
2.	Width (mm)	:	1130
3.	Height (mm)	:	1588
4.	Weight (kg) without engine	:	190
F.	Average component Shaft Speeds (with load)		
1.	Threshing cylinder (rpm)	:	600
2.	Fan (rpm)	:	2682
G.	Threshing cylinder		
1.	Type	:	Open cylinder
2.	Size, L x D (mm)	:	710 x 506
3.	Cylinder teeth		
a)	Type	:	Pegteeth
b)	Size, L x D (mm)	:	75 x 11
c)	No. per anchor bar	:	7
d)	Means of attachment	:	Bolted
e)	Arrangement	:	Offset with respect to the pegteeth in the next anchor bar
f)	Material	:	Mild steel

Table 1, Detailed specifications of the RJR-AGRO MT101311 Thresher (Cont'n)

4.	No. of anchor bars	:	8
5.	Cylinder cover material	:	B.I. Sheet, 1.5 mm thick
H.	Louvers		
1.	Material	:	B.I. Sheet, 4 mm thick
2.	Number	:	6
I.	Lower concave		
1.	Grills	:	Flat Bar, 12 mm x 6 mm
2.	Spacing between grills (mm)	:	25
3.	Clearance between concave and pegteeth (mm)	:	5
J.	Fan		
1.	Type	:	Radial centrifugal
2.	No. of unit	:	1
3.	Length (mm)	:	584
4.	Number of blades	:	4
5.	Material	:	B. I. Sheet, 1.5 mm thick (housing and blade)
K.	Feeding Table		
1.	Dimensions, L x W (mm)	:	1240 x 620
2.	Height from the ground (mm)	:	1320
3.	Mode of attachment	:	Bolted
4.	Material	:	B. I. Sheet, 1.5 mm thick
L.	Main Frame	:	Hollow square bar

Table 1, Detailed specifications of the RJR-AGRO MT101311 Thresher (Cont'n)

M.	Power Transmission System		
1.	Engine to Threshing Cylinder	:	
	Engine pulley (stub pulley)	:	102 mm x 2B
	Threshing Cylinder pulley	:	305 mm x 2B x 29 mm
2.	Threshing Cylinder to Fan	:	
	Threshing Cylinder pulley	:	254 mm x 1B x 29 mm
	Fan pulley	:	51 mm x 1B x 25 mm
N.	Transport Device	:	Manually carried
O.	Discharge device	:	None
P.	Engine (Prime mover)		
1.	Brand	:	KUBOTA
2.	Model	:	RK80-1T
3.	Make	:	Indonesia
4.	Rated maximum power (kW)	:	5.97 @ 2400 rpm
5.	Rated continuous power (kW)	:	5.22 @ 2200 rpm
6.	Type	:	Four-stroke, horizontal single cylinder diesel engine
7.	Starting system	:	Manual cranking
8.	Cooling system	:	Water-cooled
9.	Weight (kg)	:	88.0

RESULTS AND DISCUSSIONS

Results of field performance tests are shown in **Table 2 - 5**. Refer to **Annex 1** for formula used in the computation of results.

Threshing Capacity

Test results showed that the machine had a Threshing Capacity of 1024 kg/h using an unknown rice variety at 18.2% MC_{wb}, Grain-Straw Ratio of 0.62, and Straw Length of 539 mm.

For purposes of standardization, the Corrected Threshing Capacity based on 20% MC_{wb}, 0.55 Grain-Straw Ratio, and 100% Purity, was 877.4 kg/h.

Performance

The performance of the thresher was evaluated in terms of its Threshing Recovery, Threshing Efficiency, Threshing Losses, Purity and Grain Quality.

Threshing Efficiency and Recovery. The Threshing Efficiency of the machine was 100% and the Threshing Recovery was 99.71%

Threshing Loss. Total Threshing Loss was 0.29% consisting of 0.12% Blower Loss, 0.17% Separation Loss, no Unthreshed Loss and no Scattering Loss.

Purity. The Purity of collected threshed grain was 94.52%.

Grain quality. Quality of threshed grain was determined in terms of percent cracked and broken grains. Results showed that the threshed output had a net percentage Cracked Grains of 4.34% and no Broken Grains. (Cracked grains referred to those grains whose endosperms were partially fractured while broken grains referred to those grains that were split into 2 or more pieces.)

Fuel Consumption

During testing, the KUBOTA RK80-1T diesel engine had a Fuel Consumption Rate of 1.57 L/h.

Labor Requirement

Six persons were needed to operate the machine (**Fig. 7**): two for hauling of test material to the thresher; one for feeding; one for collecting the clean threshed grains; and, two for bagging.

Observation:

1. The thresher's belt power transmission was not provided with belt guard (**Fig. 8**).



Figure 7. Operators of the thresher

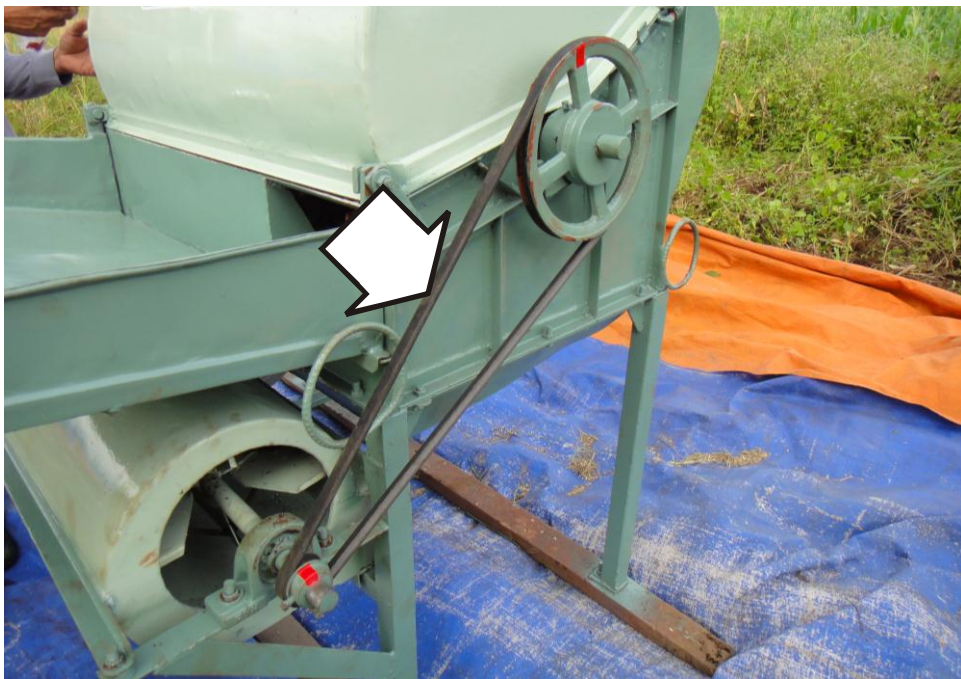


Figure 8. No belt guard

PERFORMANCE TEST

Machine Tested: RJR-AGRO Mini Rice Thresher
Model: MT101311
powered by
KUBOTA Diesel Engine
Model: RK80-1T

Location of Test: Brgy. Colonia, Valencia City

Date of Test : October 13, 2011

Table 2. Condition of crop used during the test

ITEMS	Data
1. Rice variety	Unknown
2. Days after harvest	1
3. Straw length (mm)	539
4. Grain-Straw Ratio	0.62
5. Grain moisture content (%), w.b.	18.2

Table 3. Speed of thresher components during the test

ITEMS	Data
1. Engine (rpm)	
Without load	2378
With load	2186
2. Threshing Cylinder (rpm)	
Without load	665
With load	600
3. Threshing Cylinder Peripheral Speed (m/min)	
Without load	1057
With load	954
4. Fan (rpm)	
Without load	2932
With load	2682

Table 4. Thresher performance during the test

ITEMS	Data
1. Thresher capacity (kg/h)	
Actual	1024
Corrected*	877.4
2. Purity of threshed grain (%)	94.52
3. Threshing recovery (%)	99.71
4. Threshing efficiency (%)	100
5. Blower loss (%)	0.12
6. Separation loss (%)	0.17
7. Unthreshed loss (%)	Nil
8. Scattering loss (%)	Nil
9. Total threshing loss (%)	0.29
10. Fan air velocity (m/s)	
Without load	9.79
11. Noise level [db(A)]	
a. Bagger's ear level	
Without load	97.5
With load	98.5
b. Feeder's ear level	
Without load	98.0
With load	99.0
12. Primemover's fuel consumption (KUBOTA RK80-1T Diesel Engine), L/h	1.57

* Adjusted to 20% grain moisture content, 0.55 grain-straw ratio and 100% purity of threshed grain

Table 5. Laboratory grain analysis of samples gathered during testing

ITEMS	Data
1. Before Threshing	
Cracked	2.33
Broken	0
2. After Threshing	
Cracked	6.67
Broken	0
3. Net Percent	
Cracked	4.34
Broken	0

DATA COMPUTATION

ANNEX 1. Formula used during calculations and testing

1. Grain-Straw Ratio (R)

$$R = \frac{W_g}{W_s} \quad \text{where:}$$

W_g = weight of grains (g)

W_s = weight of sample (grain and straw), g

2. Fuel consumption (F_C)

$$F_C = \frac{F_l}{T_o} \quad \text{where:}$$

F_C = Fuel consumption (L/h)

F_l = Total amount of fuel consumed (L)

T_o = Time of operation (h)

3. Actual Threshing Capacity (C_a)

$$C_a = \frac{W_c}{T_o} \quad \text{where:}$$

C_a = Actual threshing capacity (kg/h)

W_c = Weight of threshing output (kg)

T_o = Duration of operation (h)

4. Corrected Threshing Capacity (C_C)

$$C_C = \frac{100 - MC_o}{100 - MC_m} \times \frac{R_m}{R_o} \times P \times C_a \quad \text{where:}$$

C_C = Corrected threshing capacity (kg/h)

MC_o = Observed grain moisture content (%)

MC_m = Grain moisture content, at 20%

P = Purity (%)

R_m = Standard grain-straw ratio of 0.55

R_o = Observed grain-straw ratio

5. Purity (P)

$$P = 100\% - \% \text{Impurity}$$

6. Impurity (I)

$$I = \frac{G_{us} - G_{cs}}{G_{us}} \quad \text{where:}$$

I = Impurity (%)

G_{us} = Uncleaned grain sample (g)

G_{cs} = Cleaned grain sample (g)

7. Threshing Efficiency (T_e)

$$T_e = \frac{W_c + S_l + B_l}{W_c + S_l + B_l + U_l} \times 100 \quad \text{where:}$$

T_e = Threshing Efficiency (%)

S_l = Separation loss (kg)

B_l = Blower loss (kg)

U_l = Unthreshed loss (kg)

8. Threshing Recovery (T_r)

$$T_r = \frac{W_c}{W_c + S_l + B_l + U_l} \times 100$$

9. Percent Unthreshed loss (U_l)

$$U_l = \frac{U_l}{W_c + S_l + B_l + U_l} \times 100$$

10. Percent Separation loss (S_l)

$$S_l = \frac{S_l}{W_c + S_l + B_l + U_l} \times 100$$

11. Percent Blower loss (B_l)

$$B_l = \frac{B_l}{W_c + S_l + B_l + U_l} \times 100$$

12. Cracked grain (C_g)

$$C_g = \frac{N_{Cg}}{100 \text{ grain sample}} \times 100 \quad \text{where:}$$

C_g = Percent cracked grain

N_{Cg} = Number of cracked grain

13. Broken grain (B_g)

$$B_g = \frac{N_{Bg}}{100 \text{ gram sample}} \times 100 \quad \text{where:}$$

B_g = Percent broken grain

N_{Bg} = Weight of broken (g)

14. Net Cracked Grains

$$NG_C = IG_C - FG_C \quad \text{where;}$$

NG_C = Net Cracked Grains

IG_C = Number of Cracked grains before threshing

FG_C = Number of Cracked grains after threshing

15. Net Broken Grains

$$NG_B = IG_B - FG_B \quad \text{where;}$$

NG_B = Net Broken Grains

IG_B = Weight of Broken Grains before threshing

FG_B = Weight of Broken Grains after threshing

Test Engineer:

DARWIN C. ARANGUREN

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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